

Research Report on Entity Relationship Diagram (ERD) Attributes for an Immutable Digital Chain of Custody System

Introduction

An Entity Relationship Diagram (ERD) is a visualization of the structure of a database. This type of diagram shows entities present in the system, their attributes, and the relationships between them. An ERD is used while designing a database as the blueprint of the database which implements in such a way that all the required information will be organized in the proper way. While designing an immutable digital chain of custody system, an ERD will contain sufficient amount of information to keep the integrity of the evidence, accountability, and traceability of all actions taken with respect to the digital evidence.

Objectives of the ERD Design

The general objective of this Entity Relationship Diagram (ERD) design is to create an immutable database design for a digital chain of custody system. In this case, the database design will ensure safe management of digital evidence by maintaining the integrity, accountability, and traceability of the evidence.

Specific objectives include:

1. Entity identification.
2. Attribute definition for entities.
3. Entities' relationships.
4. Primary and foreign key definitions.
5. Traceability and legal admissibility of evidence.
6. Immutability via audit logging and custody.

Entities Identification

Identification of Entities

The following are the six main entities that will make up the proposed digital chain of custody system

User

This entity holds the details of users of the system, which include forensic experts, investigators, custodians, and other authorized personnel.

Case

This entity holds details about the cases being investigated and is the parent entity of all evidence.

Evidence

This entity holds the evidence and their metadata like the integrity check hash values.

Custody Record

This entity holds the history of all transfers and handling of evidence.

Audit Log

This entity keeps records of all user activities done using the system.

Location

This entity holds details about physical or digital storage locations for evidence.

Attributes Identification for Key Entities and their descriptions

User Entity

UserID : Unique Identifier

FullName : Full name of User

Email : Email of User

Role : Role of User (Example: Investigator, Analyst, Admin)

Department : Department/Unit of User

PhoneNumber : User's contact number

DigitalSignature : Digital Signature of User

Status : Status of account

Case Entity

CaseID : Unique Case Identifier

UserID : Investigator Assigned to the case

CaseTitle : Title of Case

CaseNumber : Official Case Number

CaseDescription : Description of Investigation
InvestigationType : Type of Investigation
DateOpened : Date on which case was opened
Status : Status of Case

Evidence Entity

EvidenceID : Unique Identifier for evidence
CaseID : Case for which evidence is associated
LocationID : Location where evidence is stored
EvidenceName : Name of Evidence
EvidenceType : Type of Evidence
Description : Description of Evidence
DeviceType : Source Device Type
CollectionDate : Date of Evidence Collection
CollectionLocation : Collection Location of Evidence
HashAlgorithm : Hash Algorithm used
HashValue : Hash Value generated
FileSize : Size of evidence file
FileFormat : Format of Evidence file
CurrentStatus : Current Status of Evidence

Custody Record Entity

CustodyID : Unique ID for Custody Transaction
EvidenceID : ID of Evidence Associated
CustodianID : ID of Custodian
TransferDate : Date of Transfer
TransferTime : Time of Transfer
TransferReason : Reason of Transfer
FromUserID : ID of User who transferred the Evidence
ToUserID : ID of User to whom the evidence was transferred
Signature : Signature of Custodian
VerificationStatus : Verification result

Audit Log Entity

LogID : Unique log identifier
UserID : User performing action
ActionPerformed : Action executed
Timestamp : Date and time of action
IPAddress : User IP
DeviceInformation : Device information

Result : Outcome of action

Location Entity

LocationID : Unique location identifier

LocationName : Name of storage location

Address : Name of storage location

RoomNumber : Room or compartment number

StorageType : Type of storage facility

Primary Keys

Entity	Primary Key
User	UserID
Case	CaseID
Evidence	EvidenceID
Custody Record	CustodyID
Audit Log	LogID
Location	LocationID

Foreign Keys

Entity	Foreign Key	References
Case	UserID	User
Evidence	CaseID	Case
Evidence	LocationID	Location
Custody Record	EvidenceID	Evidence
Custody Record	ToUserID	User
Custody Record	FromUserID	User
Audit Log	UserID	User
Audit Log	EvidenceID	Evidence

Relationships and Cardinalities

Relationship	Cardinality
User manages Case	1:M
Case contains Evidence	1:M
Evidence generates Custody Records	1:M
User handles Custody Records	1:M
User creates Audit Logs	1:M
Location stores Evidence	1:M

Explanation of the relationship and cardinalities

User and Case : A single user may manage multiple cases. However, each case is assigned to one responsible user.

Case and Evidence : A case may contain multiple evidence items. Each evidence item belongs to a single case.

Evidence and Custody Record : An evidence item may generate multiple custody records throughout its lifecycle. Each custody record relates to one evidence item.

User and Custody Record : A user may participate in multiple custody transactions. Each custody record is associated with one custodian.

User and Audit Log : A user may perform many actions within the system. Each action generates a corresponding audit log record.

Location and Evidence : A storage location may contain multiple evidence items. Each evidence item is stored in one location at a given time.

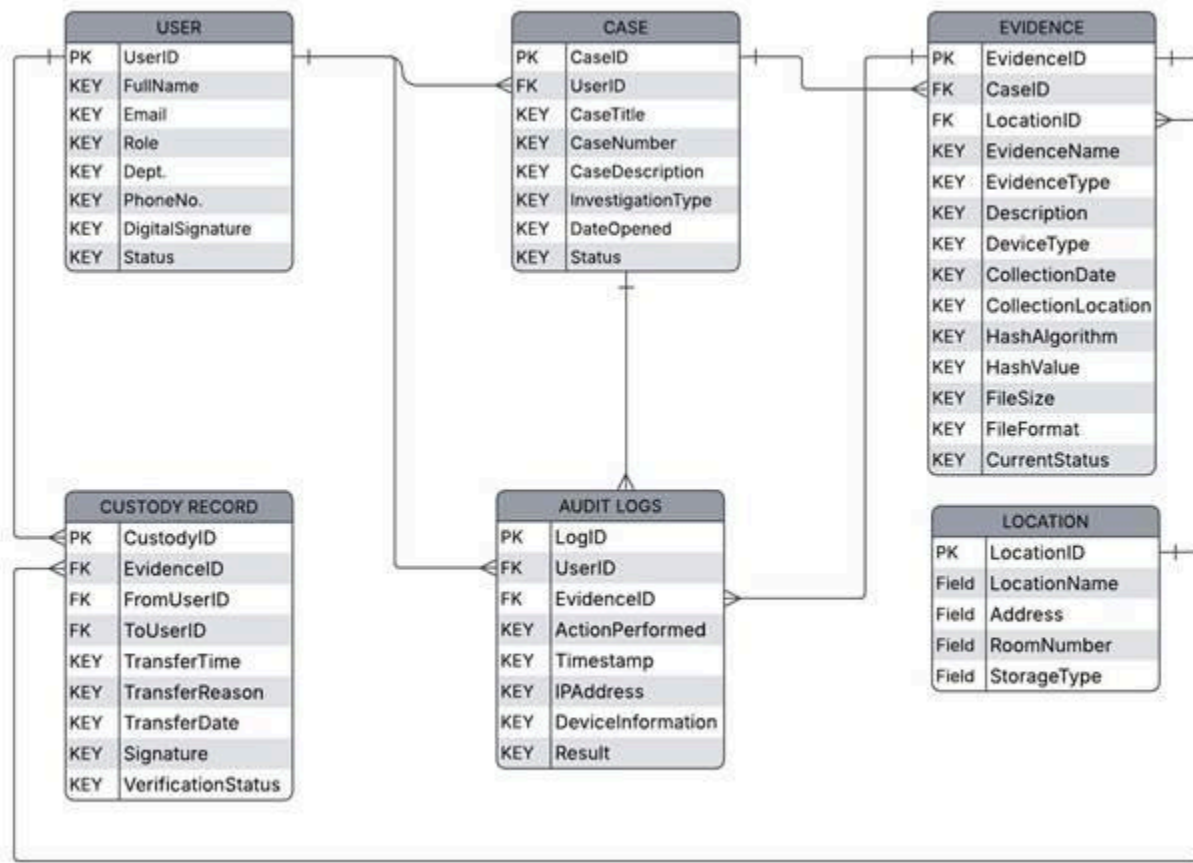
Business Rules

Business rules define the constraints and requirements that govern how data is created, managed, and related within a system. They help establish entity relationships and cardinalities in the ERD while ensuring data integrity and consistency. In this chain-of-custody system, business rules ensure proper evidence handling, accountability, and compliance with forensic and legal requirements.

The following business rules have been identified for the proposed system:

1. A user may manage multiple cases.
2. Each case must be assigned to one user.
3. A case may contain multiple evidence items.
4. Each evidence item belongs to one case.
5. Each evidence item must have a unique EvidenceID.
6. An evidence item may have multiple custody records.
7. Every custody record must reference one evidence item.
8. Every custody record must identify one sending user (FromUserID).
9. Every custody record must identify one receiving user (ToUserID).
10. A user may participate in multiple custody transactions.
11. A user may generate multiple audit logs.
12. Every audit log must be associated with one user.
13. A location may store multiple evidence items.
14. Every evidence item must be associated with one storage location.
15. Every user must possess a unique UserID.

ERD DIAGRAM



Design for Immutability

As per the above proposed ERD, the design supports the chain-of-custody process for immutability.

Each evidence transfer takes place through the Custody Record entity by means of attributes named FromUserID and ToUserID. Each transfer creates a new custody record, which contains all information about who transferred the evidence, who received it, date of transfer, time and purpose of the transfer.

Audit Log entity keeps all information regarding the user activities and is immutable. The audit logs are appended and never changed after creation.

Evidence integrity is ensured by the use of the cryptographic hashes contained within the Evidence entity. Hashes allow investigators to verify whether there has been any alteration of the evidence during storage, transfer and analysis.

Therefore, the Custody Record and Audit Log entities ensure a permanent record for the legal admissibility of the evidence and forensic integrity. They comply with the requirements set out by ISO/IEC 27037.

Normalization Considerations

Following normalization principles the design was done to reduce the level of data redundancy and improve data integrity. Users, Cases, Evidence, Custody Record, Audit Logs and Storage Location entities have been developed. All relationships take place through foreign keys.

The proposed design fulfills Third Normal Form (3NF).

*Third Normal Form (3NF) is a database design rule that stops data copy and fixes errors. To reach 3NF, a table must be in [Second Normal Form \(2NF\)](#) and have **no transitive dependencies**. [[1](#), [2](#), [3](#), [4](#)]*

<https://www.freecodecamp.org/news/database-normalization-1nf-2nf-3nf-table-examples/>

Conclusion

In this report, the metadata attributes and the ER diagram have been developed for the immutable digital chain of custody system. These entities, attributes, relationships, and business rules give a structured design for the management of the digital evidence in a secure manner. The use of custody records, audit logs, and hash values is essential to ensure the integrity of the evidence in the investigative process. This design is therefore well suited for the development of the chain of custody management system.